

The Closing Web

Consent, Copyright, and the Crisis in AI Training Data

Audit synthesis

Cooper et al. (2025), *Review of Network Economics* | Yang & Zhang (2024), *arXiv:2402.17801v2*
Fletcher (2024), *Reuters Institute Factsheet* | Kim et al. (2025), *IMC '25* | Longpre et al. (2025),
DPI / MIT

25.9%

of OpenAI's head-domain training tokens
were withdrawn via robots.txt — in twelve months.

The web is closing faster than any compute curve.

Longpre et al. (2025), Table 3 (HEAD_C4); restriction rate, mid-2023 to April 2024.

AI training rests on four pillars — all four are cracking

Pillar	What it does	Status (this deck)
Consent signal	Tells crawlers what is permitted	Eroding (Longpre 2025)
Voluntary compliance	Bots honor the signal	Selective (Kim 2025)
Premium supply	Provides quality training input	Withdrawing (Fletcher 2024)
Legal backstop	Enforces when norms fail	Fragmenting (Cooper; Yang)

In one year, head-domain training tokens went from 3% restricted to 33%

Corpus slice	Mid-2023	April 2024
C4 / RefinedWeb / Dolma (full)	~1%	5–7%
HEAD_All (top 2K domains)	<3%	20–33%
SARIMA forecast, April 2025	—	+2–4% (full); +7–11% (head)

Terms of Service now restrict **45–55%** of all corpus tokens — a 26–53% relative jump in one year.

Longpre et al. (2025), Figs. 1–2; pp. 5–7. Audit of C4, RefinedWeb, Dolma.

The closing targets one crawler $26\times$ more than another

Crawler / organization	Head tokens restricted
OpenAI (GPTBot)	25.9%
Anthropic (ClaudeBot)	13.3%
Common Crawl (CCBot)	13.3%
Google-Extended (Google AI)	9.8%
Cohere	4.9%
Meta (FacebookBot)	4.1%
Internet Archive	3.2%
Google Search (Googlebot)	1.0%

Longpre et al. (2025), Table 3 (HEAD_C4). Restrictions are AI-specific, not anti-crawl.

But the signal is unenforceable

robots.txt is a polite request, not a wall.

Bot compliance *halves* as robots.txt directives become stricter

Directive	Weighted compliance
Crawl delay (30s)	0.609
Endpoint access restriction	0.310
Disallow all	0.307

The protocol fails on contact

The advisory directive (delay) is followed twice as often as the exclusionary one. Switching from “be polite” to “stay out” loses half the leverage.

Kim et al. (2025), Table 5; 3.9M requests, 36 sites, 8 weeks.

Bots' stated policy is uncorrelated with their observed behavior

Bot	Claimed	Observed compliance
ClaudeBot, GPTBot, Amazonbot	respects	1.000 (disallow-all)
PerplexityBot	refuses	0.897 (endpoint)
Applebot	respects	0.043 (disallow-all)
Bytespider	refuses	0.000 (endpoint)
BrightEdge	respects	0.000 (disallow-all)

Selection effect: legitimate frontier labs are most exposed to opt-outs; lower-reputation actors are not.

Kim et al. (2025), Table 6 & Fig. 9.

Premium supply retreats first

The most valuable content leaves before the average page does.

By end of 2023, 48% of top news sites blocked OpenAI — and none reversed

Outlet type	Blocking OpenAI	Blocking Google AI
Legacy print	57%	32%
Broadcast	48%	19%
Digital-born	31%	17%

- USA leads at 79% OpenAI blocking; Spain, Mexico, Poland trail at 20–27%.
- Across 150 outlets in 10 countries, **no site unblocked** either crawler in 2023.
- Higher-reach outlets block more — *quality* retreats faster than quantity.

Fletcher (2024), Figs. 1, 3, 4. $N = 150$ Wayback-Machine longitudinal sample.

The most-restricted corpus is the corpus AI users barely query

Content category	HEAD_C4 share	ChatGPT query share
News & periodicals	~40%	<1%
Creative composition	small	~28%
Sexual roleplay	<1%	12%

The double irony

News publishers are blocking a corpus AI rarely uses at inference; AI usage concentrates in domains for which little permitted training data exists.

Longpre et al. (2025), Fig. 4. WildChat $N=1\text{M}$ conversations.

The legal backstop fragments

Three jurisdictions, four rulings, no shared doctrine.

\$1.5B already paid for the input stage — and the doctrine is still unsettled

Bartz v. Anthropic (<i>Alsup</i>)	Initial book downloads infringing → \$1.5B (largest in history); training itself ruled fair use.
Kadrey v. Meta (<i>Chhabria</i>)	Fair use <i>only because</i> “plaintiffs made the wrong arguments”; judge believes training is generally illegal .
USCO (<i>2025b</i>)	Doubts fair use applies when outputs compete in markets for training works.
LAION (<i>Hamburg</i>)	EU TDM Art. 4: ToS opt-outs likely enforceable.

Cooper et al. (2025), pp. 165–166. Two-thirds of public training datasets carry wrong/missing licenses.

Theory: naive responses backfire

The obvious fix — “just allow more training” — can destroy welfare.

When data is scarce, generous fair use **harms** AI development

- **Generous fair use** ($f \rightarrow 0$): kills creator incentives $\rightarrow$ **no human content** for training $\rightarrow$ model improvement stalls. (*Prop. 10*)
- **Strict fair use** (f too large): the AI company **abandons model improvement** entirely. (*Prop. 11*)
- **A moderate** f maximizes model quality X_2 , social welfare, and consumer surplus.

The dynamic problem

A copyright doctrine welfare-optimal in 2020 (data-abundant) becomes welfare-destroying once pre-existing data is exhausted (data-scarce, $Q_0=0$) — the regime Longpre et al. document we are entering.

Yang & Zhang (2024), Propositions 10–11; Figure 7.

Strong AI-copyrightability flips fair use from helpful to harmful

AI-copyrightability ϕ	Effect of generous fair use on AI development
Weak	Promotes development creators still produce human content for training
Strong	Hinders development too few human creators $\rightarrow$ training data dries up

X_2 is **inverted-U** in ϕ when data is scarce (*Prop. 12*). The two policy levers cannot be designed independently.

Yang & Zhang (2024), Figure 9 & Corollary 2.

A skeptic would say the closing web is rhetorical, not real

The objection. If half the bots ignore strict directives (Kim), if robots.txt is not legally binding, if frontier labs cut bilateral licenses, then “the web is closing” is a story about polite, compliant labs — not about training in practice.

The response.

- **Compliant labs are the regulated mainstream:** ClaudeBot, GPTBot and Amazonbot all hit **1.000** on disallow-all (Kim Table 6). Restrictions therefore bite hardest on the firms regulators can actually reach.
- **Spoofing is small but selective:** $<0.1\%$ of traffic is spoofed, yet spoofed bots show “little respect for robots.txt” (Kim Table 9, Fig. 11) — a perverse-incentive layer above the consent system.
- **The four pillars are joint constraints:** even if signaling and compliance fail, the supply (Fletcher) and legal (Cooper, Yang) pillars still bind. Ignoring robots.txt does not make a \$1.5B recovery go away.

Restoring a functioning data commons requires aligning four levers

Lever	Source	Concrete reform
Consent signaling	Longpre; Kim	Successor to REP-1995: machine-readable, granular, use-specific
Compliance	Kim	Legally binding obligations on identified crawlers
Premium supply	Fletcher; Cooper	Collective licensing / structured remuneration
Joint copyright design	Yang; Cooper	Set fair use f and AI- copyrightability ϕ jointly per data regime

Each lever necessary; none sufficient on its own.

The web is not running out of data.

It is running out of **consent**.